

8086 Segmentation Problems — Step-by-step Solutions

Problem 1: Find the physical address of 0A51:CD90h

Step 1: Multiply the segment by 10h (shift left by 4 bits = multiply by 16):

$$0A51h \times 10h = 0A510h$$

Add aligned hex (right-to-left), showing carries (carry after adding each column):

$$\begin{array}{r} 0A510 \text{ (0A510)} \\ + 0CD90 \text{ (0CD90)} \\ \hline \end{array}$$

Carries (per column, left to right): 0 1 1 0 0

Result digits (left to right): 172A0

Final physical address = 172A0h

Answer Problem 1: Physical address = **172A0h**

Problem 2: A memory location has physical address 4A37Bh. Compute the offset if segment number is 40FFh.

Step 1: Multiply the segment by 10h:

$$40FFh \times 10h = 40FF0h$$

Subtract (physical - segment×10h), showing borrows (per column left-to-right):

$$\begin{array}{r} 4A37B \text{ (4A37B)} \\ - 40FF0 \text{ (40FF0)} \\ \hline \end{array}$$

Borrows (per column, left to right): 0 0 1 1 0

Result digits (left to right): 0938B

Offset = 0938Bh

Answer Problem 2: Offset = **0938Bh**

Problem 3: Given the same physical address 4A37Bh, compute the segment if offset = 123Bh.

Method: Segment = (Physical Address - Offset) / 10h. (Divide result by 16)

Step 1: Subtract physical - offset:

$$\begin{array}{r} 4A37B \text{ (4A37B)} \\ - 0123B \text{ (0123B)} \\ \hline \end{array}$$

Borrows (per column, left to right): 0 0 0 0 0

Result digits (left to right): 49140

Step 2: Divide 49140h by 10h (i.e., shift right one hex digit): Result = 4914h

Answer Problem 3: Segment = **4914h**

Summary of Answers

- 1) Physical address of 0A51:CD90h = **172A0h**
- 2) Offset when physical = 4A37Bh and segment = 40FFh: **0938Bh**
- 3) Segment when physical = 4A37Bh and offset = 123Bh: **4914h**